

44
YEARS
OF EXCELLENCE

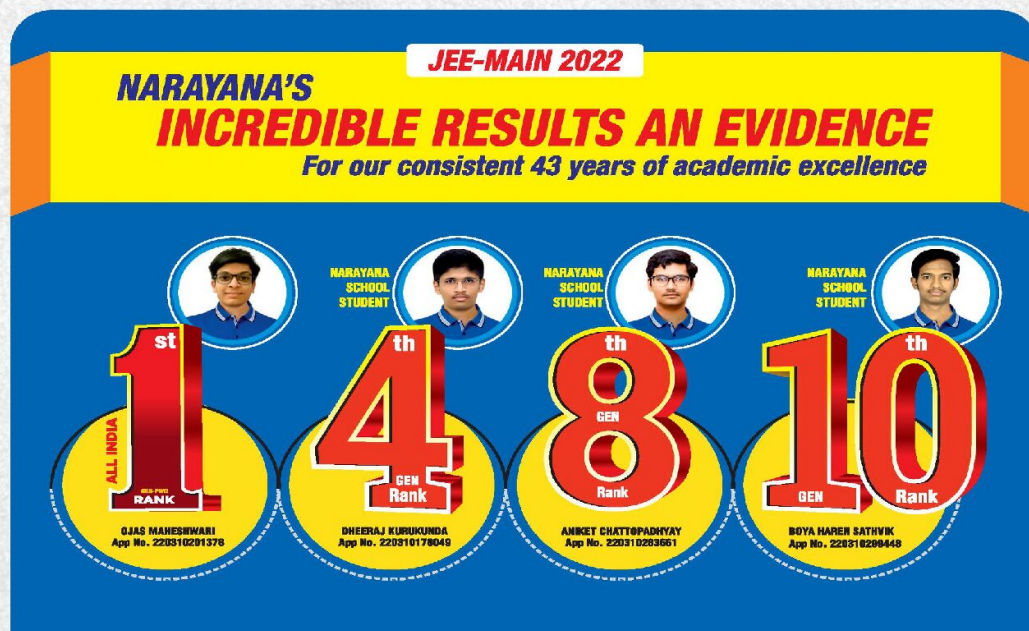


NARAYANA
IIT-JEE ACADEMY - INDIA

NARAYANA GRABS THE LION'S SHARE IN JEE-ADV.2022
EXCELLENCE IS OUR TRADITION 3402 RANKS OVERALL!



ORGANIC CHEMISTRY PART-I (JR) ADVANCED MODEL QUESTIONS



PHYSICS

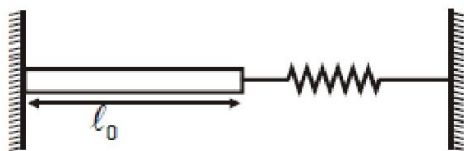
SEMI CONDUCTORS&PRACTICAL PHYSICS, PROPERTIES OF MATTER
JEE ADVANCED IMPORTANT QUESTIONS

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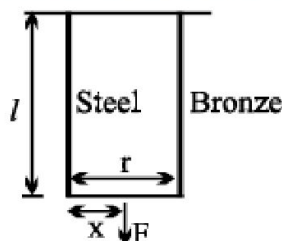


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1. A rod of mass M , area of cross section A and length ℓ_0 is connected with a spring as shown in figure. If coefficient of linear expansion of rod is α and initially no extension was there in the spring of spring constant k , then the stress developed in rod when its temperature is increased by ΔT , is [Young's Modulus of material of the rod is Y]



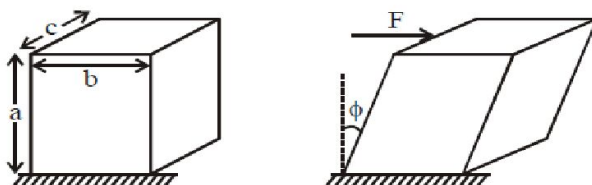
- (A) $\frac{k[Y\alpha\Delta T]\ell_0}{[k\ell_0 + YA]}$ (B) $Y\alpha\Delta T$ (C) $\left[\frac{k\ell_0 + YA}{k\ell_0}\right]Y\alpha\Delta T$ (D) $\left[\frac{k\ell_0 + Y}{k\ell_0}\right]Y\alpha\Delta T$
2. A wire of cross-section A is stretched horizontally between two clamps located $2l$ meters apart. A weight W kg is suspended from the mid point of the wire. If the mid point sags vertically through a distance $x \ll l$ the strain produced is :-
- (A) $\frac{2x^2}{l^2}$ (B) $\frac{x^2}{l^2}$ (C) $\frac{x^2}{2l^2}$ (D) none of these
3. A liquid of volumetric thermal expansion coefficient $200 \times 10^{-6} / ^\circ C$ and bulk modulus $= 1.2 \times 10^9$ Pa is filled in a spherical tank of negligible heat expansion coefficient. Its radius is 25 cm and wall thickness is 2 mm. When the temperature of the liquid is raised by $20^\circ C$, find the tensile stress developed (in MPa) in the wall of the tank ?
4. Two vertical wires each $l = 150$ cm long one of steel having cross-sectional area $A_s = 2 \times 10^{-6} m^2$ and second of bronze of cross-sectional area $A_b = 1 \times 10^{-6} m^2$ are suspended from a ceiling with horizontal distance $r = 55$ cm as shown in figure. A light but rigid horizontal rod connect lower ends of wires. Where should a vertical downward force F be applied on the rod such that it remains horizontal even after application of force. Given : for steel $Y_s = 2 \times 10^{11} N / m^2$, for bronze $Y_b = 1.5 \times 10^{11} N / m^2$.



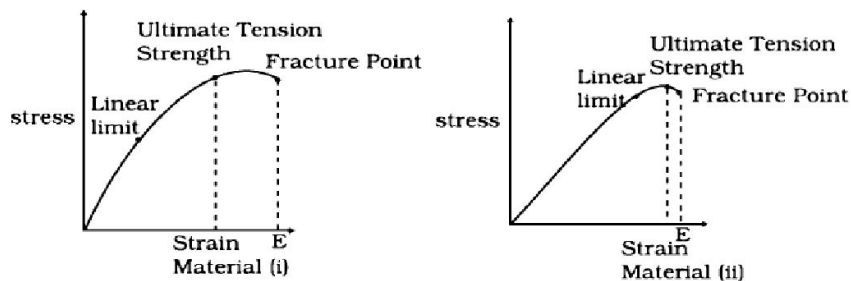
5. Two identical steel cubes each of mass 50 g & side 1 cm collide head-on face to face with a speed of 10 cm/s each. Find the maximum compression (in nm) of each. [Young's modulus for steel $2 \times 10^{11} N / m^2$]

6. A thin ring of radius $R = 2$ m made of material of density $\rho = 10^4 \text{ kg/m}^3$ and Young modulus $Y = 2 \times 10^{11} \text{ N/m}^2$ lies on a smooth horizontal table. If the ring is rotated about its centre in its own plane with angular velocity $\omega = 10 \text{ rad/s}$. Calculate the small change in radius. (In multiple of 10^{-5} m)
7. A steel cylindrical rod of length l and radius r is suspended by its end from the ceiling. Young's modulus of steel is Y . Density of steel is ρ . The elastic deformation energy of the rod is U . The value of $\pi \rho^2 l^3 r^2 g^2$
8. We would like to increase the length of a 15 cm long copper rod of cross-section 4 mm^2 by 1 mm. The energy absorbed by the rod if it is heated is E_1 . The energy absorbed by the rod if it is stretched slowly is E_2 . Then find E_1 / E_2 . If your answer is x fill value of $x/100$.
[Various parameters of Copper are: Density = $9 \times 10^3 \text{ kg/m}^3$, Thermal coefficient of linear expansion = $8 \times 10^{-5} \text{ K}^{-1}$, Young's modulus = $135 \times 10^9 \text{ Pa}$, Specific heat = 400 J/kg-K]
9. Overall changes in volume and radii of a uniform cylindrical steel wire are 0.2% and 0.002% respectively when subjected to some suitable force. Longitudinal tensile stress acting on the wire is :-
($Y = 2.0 \times 10^{11} \text{ Nm}^{-2}$)
(A) $3.2 \times 10^9 \text{ Nm}^{-2}$ (B) $3.2 \times 10^7 \text{ Nm}^{-2}$ (C) $3.6 \times 10^9 \text{ Nm}^{-2}$ (D) $3.9 \times 10^8 \text{ Nm}^{-2}$
10. A solid sphere of radius R made of material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid. When a mass m is placed on the piston to compress the liquid, the fractional change in the radius of the sphere $\delta R / R$ is
(A) mg/AK (B) $mg/3AK$ (C) mg/A (D) $mg/3AR$
11. A wire suspended vertically from one of its ends is stretched by attaching a weight of 200 N to the lower ends. The weight stretches the wire by 1 mm. Then the elastic energy stored in the wire is -
(A) 0.2 J (B) 10 J (C) 20 J (D) 0.1 J

12. A cuboidal block of sides a , b & c is fixed on ground. The top is pushed by a horizontal force F as shown. The angle ϕ by which the block deforms is : (η is modulus of rigidity)



- (A) $\frac{F}{ab\eta}$ (B) $\frac{F}{ac\eta}$ (C) $\frac{F}{bc\eta}$ (D) $\frac{F}{\sqrt{b^2 + c^2}\eta}$
13. A wire is suspended from the ceiling and stretched under the action of a weight F suspended from its other end. The force exerted by the ceiling on it is equal and opposite to the weight.
- (A) Tensile stress at any cross section A of the wire is F/A .
 (B) Tensile stress at any cross section is zero.
 (C) Tensile stress at any cross section A of the wire is $2F/A$.
 (D) Tension at any cross section A of the wire is F .
14. A copper and a steel wire of the same diameter are connected end to end. A deforming force F is applied to this composite wire which causes a total elongation of 1 cm. The two wires will have
- (A) the same stress. (B) different stress. (C) the same strain. (D) different strain.
15. A body of mass M is attached to the lower end of a metal wire, whose upper end is fixed. The elongation of the wire is l .
- (A) Loss in gravitational potential energy of M is Mgl
 (B) The elastic potential energy stored in the wire is Mgl
 (C) The elastic potential energy stored in the wire is $1/2 Mgl$
 (D) Heat produced is $1/2 Mgl$.
16. The stress-strain graphs for two materials are shown in figure (assume same scale).

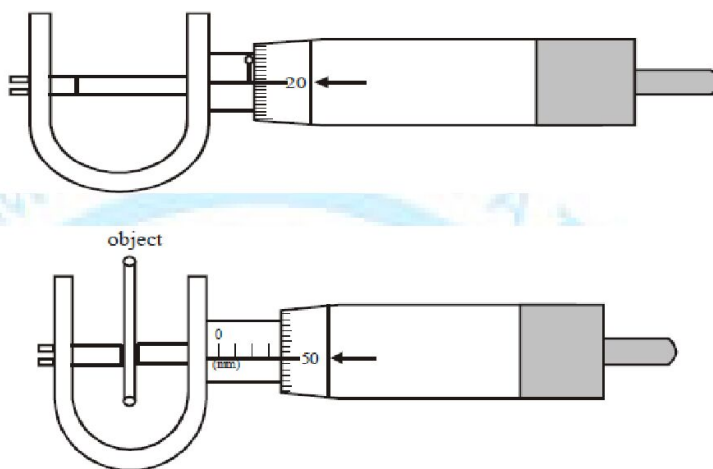


- (A) Material (ii) is more elastic than material (i) and hence material (ii) is more brittle.
- (B) Material (i) and (ii) have the same elasticity and the same brittleness.
- (C) Material (ii) is elastic over a larger region of strain as compared to (i).
- (D) Material (ii) is more brittle than material (i).
17. A composite rod consists of a steel rod of length 25 cm and area $2A$ and a copper rod of length 50 cm and area A . The composite rod is subjected to an axial load F . If the Young's modulus of steel and copper are in the ratio 2 : 1.
- (A) the extension produced in copper rod will be more.
- (B) the extension in copper and steel parts will be in the ratio 2 : 1.
- (C) the stress applied to the copper rod will be more.
- (D) no extension will be produced in the steel rod.
18. The wires A and B shown in the figure are made of the same material and have radii r_A and r_B respectively. The block between them has a mass m . When the force F is $mg/3$, one of the wires
-
- (A) A breaks if $r_A = r_B$
- (B) A breaks if $r_A < 2r_B$
- (C) Either A or B may break if $r_A = 2r_B$
- (D) The lengths of A and B must be known to predict which wire will break
19. In Post office box experiment, to avoid misleading deflection in galvanometer due to self induced current [back emf/back current] one should press
- (A) both the cell key & galvanometer key simultaneously
- (B) first cell key & then galvanometer key with delay of 2 to 3 second

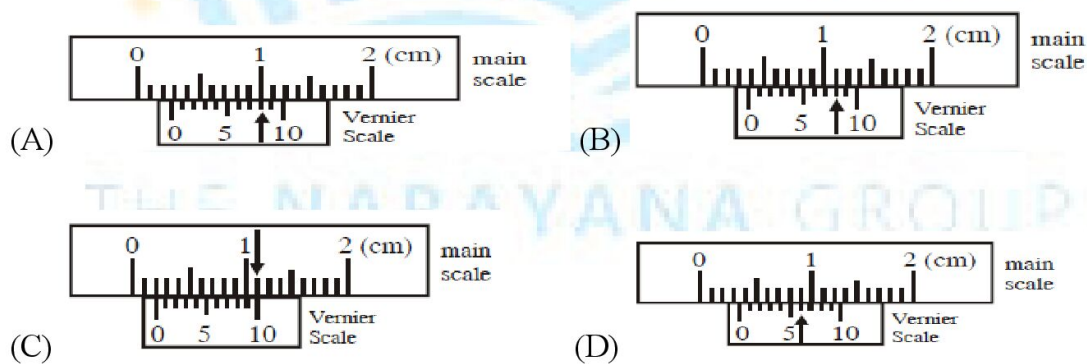
(C) first galvanometer key & then cell key with delay of 2 to 3 second

(D) any of the manner mentioned above

20. When a piece of wire is held diametrically in a screw gauge [pitch = 1 mm, number of division on circular scale = 100]. The readings obtained are as shown:



Now if we measure the same with help of vernier callipers [1 MSD = 1mm, 10 divisions of vernier coinciding with 9 divisions of main scale] having a negative zero error of 0.5 mm, then find which of the following figures correctly represents the reading.



21. Column I shows various practical errors/corrections and column II shows the experimental setups. Choose the correct matching.

1. Radiation correction

(i) Optical bench

2. End correction

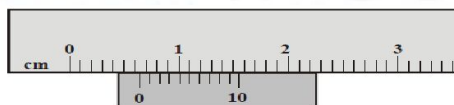
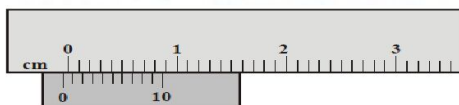
(ii) Joule's calorimeter

3. Backlash error (iii) Searle's apparatus
4. Index error (iv) Meter bridge
- (A) 1.(iii), 2. (i), 3. (iv), 4. (ii) (B) 1. (ii), 2. (iv), 3. (i), 4. (iii)
- (C) 1. (ii) 2. (iv), 3. (iii), 4. (i) (D) 1. (ii), 2. (iii), 3. (i), 4. (iv)
22. If $a = 8 \pm 0.08$ and $b = 6 \pm 0.06$, Let $x = a + b, y = a - b, z = a \times b$. The correct order of % error in x, y and z is
- (A) $x = y < z$ (B) $x = y > z$ (C) $x < z < y$ (D) $x < z < y$
23. In a Searle's experiment for determination of Young's Modulus, when a load of 50 kg is added to a 3 meter long wire micrometer screw having pitch 1 mm needs to be given a quarter turn in order to restore the horizontal position of spirit level. Young's modulus of the wire if its cross sectional area is $10^{-5} m^2$ is :-
- (A) $6 \times 10^{11} N / m^2$ (B) $1.5 \times 10^{11} N / m^2$ (C) $3 \times 10^{11} N / m^2$ (D) None
24. The length of the string of a simple pendulum is measured with a meter scale to be 63.5 cm, the radius of the bob plus the hook is measured with the help of vernier caliper to be 1.55 cm. Select the incorrect statement:-
- (A) Least count of meter scale is 0.1 cm
- (B) Least count of vernier caliper is 0.01 cm
- (C) Effective length of pendulum is 65.1 cm
- (D) Effective length of pendulum is 65.2 cm
25. The length of a rectangular plate is measured by a meter scale and is found to be 10.0 cm. Its width is measured by vernier callipers as 1.00 cm. The least count of the meter scale and vernier callipers are 0.1 cm and 0.01 cm respectively (Obviously). Maximum permissible error in area measurement is :-
- (A) $\pm 0.2 cm^2$ (B) $\pm 0.1 cm^2$ (C) $\pm 0.3 cm^2$ (D) 0

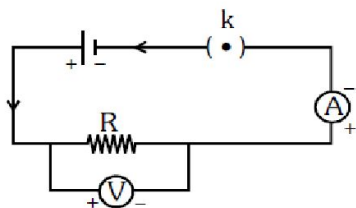
26. Match the column -I with column – II.

Column-(Number)	Column-II (Number of significant digits)
(A) 1001	(p) 5
(B) 0100.1	(q) 4
(C) 100.100	(r) 6
(D) 0.001001	(s) 3
(A) (A)-q, (B)-r, (C)-s, (D)-p	(B) (A)-q, (B)-q, (C)-r, (D)-q
(C) (A)-r, (B)-q, (C)-p, (D)-s	(D) (A)-p, (B)-q, (C)-s, (D)-r

27. The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{L/g}$. Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90 s using a wrist watch of 1s resolution. What is the accuracy in the determination of g ? If answer is n% fill 18n.
28. The pitch of a screw gauge is 1 mm and there are 50 divisions on its cap. When nothing is put in between the studs, 44th division of the circular scale coincides with the reference line and zero of main scale is not visible. When a plate is placed between the studs, the main scale reads 5 divisions and the circular scale reads 25 divisions. The true thickness of plate is a $\times 10^{-5}$ m, find the value of a.
29. The main scale of a vernier calipers reads in millimeter and its vernier is divided into 10 divisions which coincides with 9 divisions of the main scale. The reading for shown situation is found to be $\frac{23x}{10}$ mm. Find the value of x.



30. Mr. Vivek Taparia performed an experiment to verify Ohm's law. He connected following circuit to measure voltage and current.



Here R is the unknown resistance, V the voltmeter, A the ammeter and K is the key. The value of R from following readings is given by $\alpha \times 10^8 \text{ k}\Omega$, then find the value of $\alpha + \beta$.

V(volt)	1	2	3	4	5
I(mA)	1.40	2.83	5.68	7.11	8.54

31. In a certain system of units, 1 unit of time is 5 sec, 1 unit of mass is 20 kg and 1 unit of length is 10 m. In this system, one unit of power will correspond to :-
- (A) 16 watts (B) $\frac{1}{16}$ watts (C) 25 watts (D) none of these
32. If the unit of length be doubled then the numerical value of the universal gravitation constant G will become (with respect to present value)
- (A) Double (B) Half (C) 8 times (D) $1/8$ times
33. If in a system, the force of attraction between two point masses of 1 kg each situated 1 km apart is taken as a unit of force and is called notwen (newton written in reverse order) & if $G = 6.67 \times 10^{-11} \text{ N-m}^2\text{kg}^{-2}$ in SI units then which of the following is true?
- (A) 1 notwen = 6.67×10^{-11} newton (B) 1 newton = 6.67×10^{-17} notwen
- (C) 1 notwen = 6.67×10^{-17} newton (D) 1 newton = 6.67×10^{-12} notwen
34. In two different systems of units an acceleration is represented by the same number, while a velocity is represented by numbers in the ratio 1 : 3. The ratios of unit of length and time are respectively
- (A) 3, 9 (B) 9, 3 (C) 1, 1 (D) None of these

35. **Statement-1:** Whenever the unit of measurement of a quantity is changed, its numerical value changes.

And

Statement-2 : Smaller the unit of measurement smaller is its numerical value.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement -1
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.

Paragraph for Question Nos. 36 to 38

A physical quantity is a physical property of a phenomenon, body, or substance, that can be quantified by measurement.

The magnitude of the components of a vector are to be considered dimensionally distinct. For example, rather than an undifferentiated length unit L , we may represent length in the x direction as L_x , and so forth. This requirement stems ultimately from the requirement that each component of a physically meaningful equation (scalar or vector) must be dimensionally consistent. As an example, suppose we wish to calculate the drift S of a swimmer crossing a river flowing with velocity V_x and of width D and he is swimming in direction perpendicular to the river flow with velocity V_y relative to river,

Assuming no use of directed lengths, the quantities of interest are then V_x, V_y both dimensioned as $\frac{L}{T}$,

S the drift and D width of river both having dimension L . With these four quantities, we may conclude that the equation for the drift S may be written: $S \propto V_x^a V_y^b D^c$

Or dimensionally $L = \left(\frac{L}{T}\right)^{a+b} \times (L)^c$ from which we may deduce that $a+b+c=1$ and $a+b=0$, Which leaves one of these exponents undetermined. If, however, we use directed length dimensions, then V_x will be dimensioned as $\frac{L_x}{T}$, V_y as $\frac{L_y}{T}$, S as L_x and D as L_y . The dimensional equation becomes :

$L_x = \left(\frac{L_x}{T}\right)^a \left(\frac{L_y}{T}\right)^b (L_y)^c$ and we may solve completely as $a = 1$, $b = -1$ and $c = 1$. The increase in deductive power gained by the use of directed length dimensions is apparent.

36. Which of the following is not a physical quantity
 (A) Height of a boy (B) Weight of a boy
 (C) Fever of a boy (D) Speed of a running boy
37. From the concept of directed dimension what is the formula for a range (R) of a cannon ball when it is fired with vertical velocity component V_y and a horizontal velocity component V_x assuming it is fired on a flat surface. [Range also depends upon acceleration due to gravity, g and k is numerical constant]
 (A) $R = \frac{k(V_x V_y)}{g}$ (B) $R = \frac{k(V_x)^2}{g}$ (C) $R = \frac{k(V_x)^3}{V_y g}$ (D) $R = \frac{k(V_y)^3}{V_x g}$
38. A conveyer belt of width D is moving along x -axis with velocity V . A man moving with velocity U on the belt in the direction perpendicular to the belt's velocity with respect to belt wants to cross the belt. The correct expression for the drift (S) suffered by man is given by (k is numerical constant)
 (A) $S = k \frac{UD}{V}$ (B) $S = k \frac{VD}{U}$ (C) $S = k \frac{U^2 D}{V^2}$ (D) $S = k \frac{V^2 D}{U^2}$

Answer Q.39 and Q.40 and Q.41 by appropriately matching the information given in the three columns of the following table.

L , M and T are units of length, Mass and Time respectively in a system of units.

Column-1	Column-2	Column-3
(I) $L = 10 \text{ m}$	(i) $M = 100 \text{ gm}$	(P) $T = 0.1 \text{ sec}$
(II) $L = 10 \text{ cm}$	(ii) $M = 10 \text{ kg}$	(Q) $T = 10 \text{ ms}$
(III) $L = 0.1 \text{ mm}$	(iii) $M = 10 \text{ gm}$	(R) $T = 10 \text{ sec}$
(IV) $L = 1 \text{ km}$	(iv) $M = 1 \text{ tonne}$	(S) $T = 0.01 \text{ sec}$

39. In which of the following combinations unit of force is 10^6 dyne.
 (A) (IV) (i) (P) (B) (II) (iii) (S) (C) (III) (iv) (P) (D) (I) (ii) (Q)
40. In which of the following system, unit of energy is 10^9 erg
 (A) (III) (i) (S) (B) (IV) (iii) (R) (C) (II) (iv) (Q) (D) (I) (iii) (P)

41. In which of the following system, unit for coefficient of viscosity is 100 poiseuille
 (A) (III) (ii) (S) (B) (II) (i) (Q) (C) (III) (iii) (R) (D) (IV) (iv) (P)
42. In a new system of units known as RMP, length is measured in 'retem', mass is measured in 'marg' and time is measured in 'pal'.

$$100 \text{ retem} = 1.0 \text{ meter}$$

$$1.0 \text{ marg} = 10^{-3} \text{ kilogram}$$

$$10 \text{ pal} = 1.0 \text{ second}$$

In the given table some unit conversion factors are given. Suggest suitable match.

Column-I**Column-II**

- | | |
|-------------------------------------|-------------------------|
| (A) One SI unit of force | (P) 10^2 units of RMP |
| (B) One SI unit of potential energy | (Q) 10^3 units of RMP |
| (C) One SI unit of power | (R) 10^4 units of RMP |
| (D) One SI unit of momentum | (S) 10^5 units of RMP |
| | (T) 10^6 units of RMP |

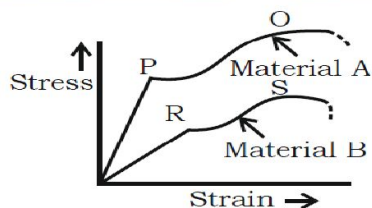
(A) (A) $\rightarrow$ (Q); (B) $\rightarrow$ (S); (C) $\rightarrow$ (R); (D) $\rightarrow$ (R)

(B) (A) $\rightarrow$ (R); (B) $\rightarrow$ (Q); (C) $\rightarrow$ (P); (D) $\rightarrow$ (R)

(C) (A) $\rightarrow$ (S); (B) $\rightarrow$ (P); (C) $\rightarrow$ (Q); (D) $\rightarrow$ (R)

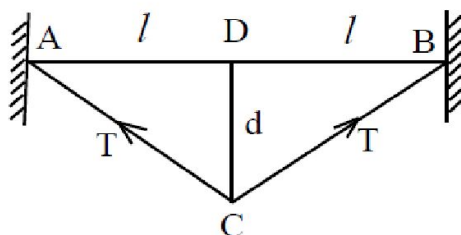
(D) (A) $\rightarrow$ (P); (B) $\rightarrow$ (R); (C) $\rightarrow$ (Q); (D) $\rightarrow$ (S)

43. The figure shows the stress-strain graphs for material A and B. From the graph it follows that



- (A) Material A has a higher Young's modulus
- (B) Material B is more ductile
- (C) Material B can withstand greater stress
- (D) Material A can withstand greater stress

44. A wire AB of length $2l$ and cross – section area a is stretched without tension between fixed points A and B. The wire is pulled at the centre into shape ACB such that $d \ll l$, the tension in the string is T and the strain produced in it is ϵ . If the Young's modulus of the material of wire is Y , then



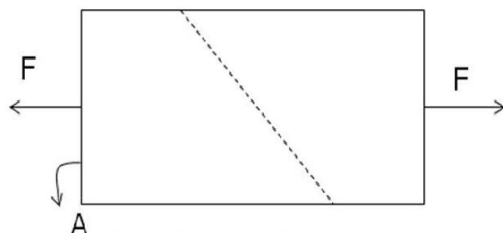
- (A) $\epsilon = \frac{d}{2l}$ (B) $\epsilon = \frac{d^2}{2l^2}$ (C) $T = aY \frac{d}{2l}$ (D) $T = aY \frac{d^2}{2l^2}$
45. A uniform rod of mass m , length l is freely suspended vertically from a fixed horizontal support.

(ρ = density of the rod, Y = Young's modulus of the rod, A = area of cross section of the rod). In this situation mark out the correct statement(s)

- (A) The elongation of the rod is $\frac{mgl}{2AY}$
- (B) The elongation of the rod is $\frac{\rho g l^2}{2Y}$
- (C) Elastic potential energy stored in the rod is $\frac{1}{6} \frac{m^2 g^2 l}{AY}$
- (D) Elastic potential energy stored in the rod is $\frac{1}{6} \frac{\rho^2 g^2 l^3 A}{Y}$
46. An elastic rod of length l , mass m Young's modulus Y and area of cross section A is rotating with angular velocity ω about its centre on a smooth horizontal surface.
- (A) The stress at a distance $\frac{l}{4}$ from the centre of rod is $\frac{3}{32} \frac{m\omega^2 l}{A}$
- (B) The elongation in the rod is $\frac{m\omega^2 l^2}{12AY}$
- (C) The elastic energy stored in the rod is $\frac{m^2 \omega^4 l^2}{488 A^2 Y}$

(D) The elastic energy stored in the rod is $\frac{m^2 \omega^4 l^2}{A^2 Y}$.

47. A bar of cross section A is subjected to equal and opposite tensile forces F at its end. Consider a plane section of the bar whose normal makes an angle θ with the horizontal axis of bar

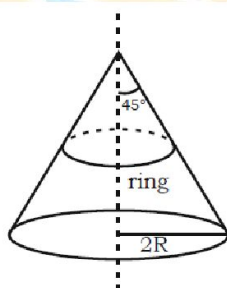


(A) The tensile stress on this plane is $\frac{F}{A} \cos^2 \theta$ (B) The shearing stress on this plane is $\frac{F}{A} \sin 2\theta$

(C) Tensile stress is maximum when $\theta = 90^\circ$

(D) Shearing stress is maximum when $\theta = 45^\circ$

48. A uniform metallic ring of mass m , radius R , cross sectional area a and Young's modulus Y is kept on a smooth cone of radius $2R$ and semi vertical angle 45° , as shown in the figure. Assume that the extension in the ring is small.



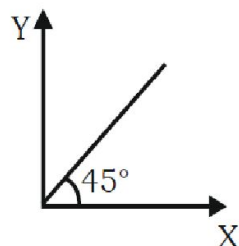
(A) The tension in the ring will be same throughout.

(B) The tension in the ring will be independent of the radius of ring (of constant mass) if radius of the ring is less than $2R$

(C) The extension in the ring will be $\frac{mgR}{aY}$

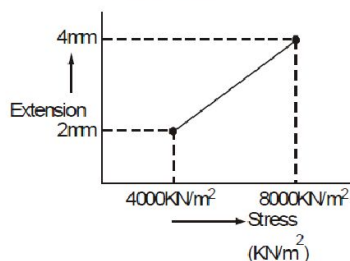
(D) Elastic potential energy stored in the ring will be $\frac{m^2 g^2 R}{8\pi Ya}$

49. The product of Young's modulus of the material of the wire with its cross-sectional area is equal to its length. Find the parameters representing X and Y-axes of the curve as shown in the figure

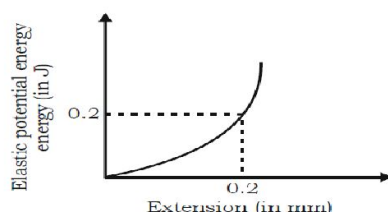


- (A) Load and increase in length
- (B) Strain energy and square of increase in length
- (C) Young's modulus and Stress
- (D) Young's modulus and strain

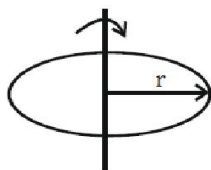
50. In determination of young modulus of elasticity of wire, a force is applied and extension is recorded. Initial length of wire is '1m'. The curve between extension and stress is depicted then young modulus of wire will be $\alpha \times 10^9 \text{ N/m}^2$. The value of ' α ' is



51. A cube of aluminum of sides 0.1 m is subjected to a shearing force of 100 N. The top face of the cube is displaced through 0.02 cm with respect to the bottom face. The shearing strain would be $10^{-3} \times K$, where the value of K is:
52. A solid sphere of radius r made of a material of bulk modulus B is surrounded by a liquid in a cylindrical container. A massless piston of area A (the area of container is also A) floats on the surface of the liquid. When a mass M is placed on the piston to compress the liquid, fractional change in radius of the sphere is $\frac{Mg}{\alpha AB}$. Find the value of α .
53. Figure shows the graph of elastic potential energy (U) stored versus extension is a parabola, for a steel wire ($Y = 2 \times 10^{11} \text{ Pa}$) of volume 200 cm^3 , the original length of the wire (in m) is



54. A thin ring made of wire of density ρ is rotated about a stationary vertical axis passing through its centre and perpendicular to the plane of the ring as shown in the figure. Determine the angular velocity (in rad /s) of ring at which the ring breaks. The wire breaks at tensile stress σ . Ignore gravity. Take $\frac{\sigma}{\rho} = 4$ and $r = 1\text{m}$.



Answer Q.55,Q.56 and Q.57 by appropriately matching the information given in the three columns of the following table.

A solid copper cylinder is placed on a horizontal plane such that flat surface of the cylinder is on the plane. Let area of flat surface of the cylinder is A , length of the cylinder is l , Young's modulus of the cylinder is Y and Poisson's ratio of the cylinder is μ , which is defined as the ratio of lateral strain to longitudinal strain. Consider lateral strain and longitudinal strains are small.

Now cylinder is subject to a vertical compressive force F directed downwards and distributed over the top flat surface.

Column-1	Column-2	Column-3
I) directly proportional to F	i) directly proportional to A	P) directly proportional to l
II) inversely proportional to F	ii) inversely proportional to A	Q) inversely proportional to l
III) $F = 1000\text{ N}$	iii) independent of A	R) independent of l
IV) $F = 2000\text{ N}$	iv) $A = 100\text{ cm}^2$	S) $l = 50\text{ cm}$

Mark the correct option for given below questions according to the data given in the above column.

55. The change in volume of the cylinder is

- (A) II, ii, P (B) I, iii, P (C) I, i, Q (D) II, i, Q

56. The fractional change in volume of the cylinder is

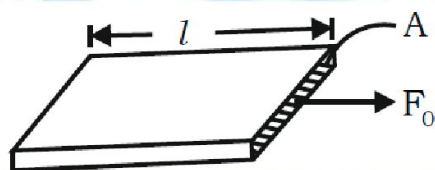
- (A) I, ii, R (B) I, iii, R (C) II, ii, P (D) II, ii, Q

57. If $Y = 100 \text{ GPa}$ and $\mu = 0.3$ for copper then the change in the volume is 2 mm^3 when

- (A) III, iii, S (B) IV, iv, S (C) III, iv, S (D) IV, iv, R

Answer Q.58, Q.59 and Q.60 by appropriately matching the information given in the three columns of the following table.

A uniform elastic plank moves due to a force which distributed uniformly over the end face on the smooth horizontal surface. The cross-sectional area is A and Young's modulus of material of plank is Y .



Column-1	Column-2	Column-3
I) directly proportional to F_0	(i) directly proportional to A	P) directly proportional to Y
II) $F_0 = 200 \text{ N}$	(ii) $A = 10 \text{ mm}^2$	Q) $Y = 3 \times 10^{10} \text{ Nm}^{-2}$
III) directly proportional to F_0^2	(iii) inversely proportional to A	R) inversely proportional to Y
IV) $F_0 = 100 \text{ N}$	(iv) $A = 5 \text{ mm}^2$	S) $Y = 2 \times 10^{10} \text{ Nm}^{-2}$

Mark the correct option for given below questions according to the data given in the above columns.

58. The strain produced in the direction of the forces is

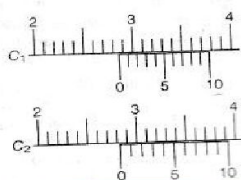
- (A) I, iv, R (B) I, iii, R (C) III, ii, R (D) III, iv, R

59. The strain energy produced per unit length of the plank is
(A) III, ii, R (B) I, ii, P (C) III, iii, R (D) I, iv, R
60. The strain produced in the direction of the forces is 0.001 for
(A) II, I, Q (B) II, iii, Q (C) IV, I, Q (D) II, iv, S
61. The de Broglie wavelength of a neutron at 927°C is λ . What will be its wavelength at 27°C ?
(A) $\frac{\lambda}{2}$ (B) λ (C) 2λ (D) 4λ
62. If an orbital electron of the hydrogen atom jumps from the ground state to a higher energy state, its orbital speed reduces to half its initial value. If the radius of the electron orbit in the ground state is r , then the radius of the new orbit would be
(A) $2r$ (B) $4r$ (C) $8r$ (D) $16r$
63. The slope of the graph of the frequency of incident light (on x-axis) versus the stopping potential (on y-axis) for a given metallic surface is
(A) h (B) $\frac{h}{e}$ (C) $\frac{e}{h}$ (D) eh
64. The radius of hydrogen atom in the ground state is $0.53\text{ \AA}$. After collision with an electron, it is found to have a radius of $2.12\text{ \AA}$. What is the principal quantum number n of the final state of the atom?
(A) $n = 1$ (B) $n = 2$ (C) $n = 3$ (D) $n = 4$
65. A 1 MeV positron and a 1 MeV electron meet each moving in opposite directions. They annihilate each other emitting two photons. If the rest mass energy of an electron is 0.51 MeV , the wavelength of each photon is
(A) $5.1 \times 10^{-3}\text{ \AA}$ (B) $10.2 \times 10^{-3}\text{ \AA}$ (C) $8.2 \times 10^{-3}\text{ \AA}$ (D) $6.2 \times 10^{-3}\text{ \AA}$
66. The resistance of metal is given by $V = IR$. The voltage across the resistance is $V = (8 \pm 0.5)\text{V}$ and current in the resistance is $I = (2 \pm 0.2)\text{A}$, the value of resistance can be represented as
(A) $(4 \pm 16.25\%) \Omega$ (B) $(4 \pm 25\%) \Omega$ (C) $(4 \pm 0.04\%) \Omega$ (D) $(4 \pm 1\%) \Omega$
67. Choose correct statements
A) The number of significant figures in 0.310×10^3 is three

- B) The length, breadth and thickness of a rectangular sheet of metal are 4.234 m, 1.00 m and 2.01 cm. Its volume is 0.0851034 m^3 .
- C) When 97.52 is divided by 2.54, the result is 38.4
- D) When 0.2 J is subtracted from 7.26 J, the result with correct number of significant figures is 7.1 J.
68. A resistance R of 100Ω is to be measured by using an ammeter and voltmeter in standard methodology. If the readings of ammeter and voltmeter are 12 mA and 1 V respectively. Which of the following conclusion is correct
- A) Measured value of R is 83Ω
- B) Percentage error in the measurement of R is 17%
- C) Resistance of voltmeter is 200Ω
- D) Resistance of voltmeter is 500Ω
69. In an experiment to find the relative density of a material, the weight of the body in air (W_a) is found to be $(10.0 \pm 0.1)\text{N}$ and the weight in water (W_w) is (5.0 ± 0.1) . If the relative density of the material is given by $\frac{W_a}{W_a - W_w}$. Then maximum percentage error in relative density is $k\%$. Find k
70. Consider a Vernier Callipers in which each 1 cm on the main scale is divided into 8 equal divisions and a screw gauge with 100 division on its Circular scale. In the Vernier Callipers, 5 divisions of the Vernier scale coincides with 4 divisions of main scale and in the screw gauge, one complete rotation of the circular scale moves it by two divisions on the linear scale.
- A) If the least count of screw gauge is 0.005mm, then the ratio of the pitch of screw gauge to least count of the vernier calliper is 2
- B) If the least count of screw gauge is 0.01mm, then the ratio of pitch of the screw gauge and least count of vernier calliper is 2
- C) If the Least count of screw gauge is 0.01mm then the ratio least count of the linear scale of the screw gauge to least count of vernier scale is 2
- D) If the least count of screw gauge is 0.005 mm, then the ratio of the least count of the linear scale of the screw gauge to least count of the vernier scale is 2.
71. $N+1$ equal division on the main scale of a Vernier Caliper coincide with N equal divisions on its Vernier scale. If each division on the main scale is of X units. The least count of the instrument is

- (A) $\frac{X}{N+1}$ (B) $\frac{NX}{N+1}$ (C) $\frac{X}{N}$ (D) $\frac{NX}{N-1}$

72. There are two Vernier Calipers both of which have 1 cm divided into 10 equal division on the main scale. The Vernier scale of one of the Callipers (C_1) has 10 equal divisions that correspond to 9 main scale divisions. The Vernier scale of the other Caliper (C_2) has 10 equal divisions that correspond to 11 main scale divisions. The readings of the two calipers are shown in the figure below. Select correct Readings of the two Callipers.



- A) Measured value (in cm) by caliper C_1 is 2.87
 B) Measured value(in cm) by caliper C_2 is 2.83
 C) Measured value(in cm) by caliper C_1 is 2.83
 D) Measured value(in cm) by caliper C_2 is 2.87

Paragraph for Questions 73 and 74

The pitch of a screw gauge is 1 mm and there are 50 divisions on its circular scale. When nothing is put between the jaws the zero of circular scale lies 3 divisions below the reference line, when a wire is placed between the jaws on linear scale division is clearly visible while 30 divisions on circular scale coincide with reference line.

73. The least count of screw gauge is
 (A) 0.002 mm (B) 0.02 mm (C) 0.02 cm (D) 0.002 cm
74. The error in the instrument is
 A) 0.006 mm with positive zero error.
 B) 0.06 mm with negative zero error.
 C) 0.006 mm with negative zero error.
 D) 0.06 mm with positive zero error
75. The dimensional formula for Planck's constant and angular momentum are

- (A) $[ML^2T^{-2}]$ and $[MLT^{-1}]$ (B) $[ML^2T^{-1}]$ and $[ML^2T^{-1}]$
 (C) $[ML^3T^1]$ and $[ML^2T^{-2}]$ (D) $[MLT^{-1}]$ and $[MLT^{-2}]$
76. Dimension of velocity gradient is
 (A) $[M^0L^0T^{-1}]$ (B) $[ML^{-1}T^{-1}]$ (C) $[M^0LT^{-1}]$ (D) $[ML^0T^{-1}]$
77. Which of the following is the dimension of the coefficient of friction?
 (A) $[M^2L^2T]$ (B) $[M^0L^0T^0]$ (C) $[ML^2T^{-2}]$ (D) $[M^2L^2T^{-2}]$
78. Which of the following sets have different dimensions?
 (A) Pressure, Young's modulus, Stress
 (B) Emf, Potential difference, Electric potential
 (C) Heat, Work done, Energy
 (D) Dipole moment, Electric flux, Electric field
79. The viscous force F on a sphere of radius α moving in a medium with velocity v is given by $F = 6\pi\eta\alpha v$. The dimensions of η are
 (A) $[ML^{-3}]$ (B) $[MLT^{-2}]$ (C) $[MT^{-1}]$ (D) $[ML^{-1}T^{-1}]$
80. A force is given by $F = \alpha t + bt^2$
 Where, t is the time. The dimensions of α and b are
 (A) $[MLT^{-4}]$ and $[MLT]$ (B) $[MLT^{-1}]$ and $[MLT^0]$
 (C) $[MLT^{-3}]$ and $[MLT^{-4}]$ (D) $[MLT^{-3}]$ and $[MLT^0]$
81. The physical quantity having the dimensions $[M^{-1}L^{-3}T^3A^2]$ is
 (A) resistance (B) resistivity
 (C) electrical conductivity (D) electromotive force
82. The dimensional formula for magnetic flux is
 (A) $[ML^2T^{-2}A^{-1}]$ (B) $[ML^3T^{-2}A^{-2}]$ (C) $[M^0L^{-2}T^{-2}A^{-2}]$ (D) $[ML^2T^{-1}A^2]$
83. Choose the wrong statement.

- (A) All quantities may be represented dimensionally in terms of the base quantities
 (B) A base quantity cannot be represented dimensionally in terms of the rest of the base quantities
 (C) The dimension of a base quantity in other base quantities is always zero
 (D) The dimension of a derived quantity is never zero in any base quantity
84. If unit of length and time is doubled, the numerical value of g (acceleration due to gravity) will be
 (A) doubled (B) halved (C) four times (D) same
85. Using mass (M), length (L), time (T) and current (A) as fundamental quantities, the dimension of permeability is
 (A) $[M^{-1}LT^{-2}A]$ (B) $[ML^{-2}T^{-2}A^{-1}]$ (C) $[MLT^{-2}A^{-2}]$ (D) $[MLT^{-1}A^{-1}]$
86. The equation of a wave is given by
- $$y = a \sin \omega \left(\frac{x}{v} - k \right)$$
- Where, ω is angular velocity and v is the linear velocity. The dimensions of k will be
 (A) $[T^2]$ (B) $[T^{-1}]$ (C) $[T]$ (D) $[LT]$
87. If the energy E , velocity (v) and force (F) be taken as fundamental quantities, then the dimensions of mass will be
 (A) $[Fv^{-2}]$ (B) $[Fv^{-1}]$ (C) $[Ev^{-2}]$ (D) $[Ev^2]$
88. If force F , length L and time T are taken as fundamental units, the dimensional formula for mass will be
 (A) $[FL^{-1}T^2]$ (B) $[FLT^{-2}]$ (C) $[FL^{-1}T^{-1}]$ (D) $[FL^3T^2]$
89. The ratio of the dimensions of Planck's constant and that of the moment of inertia is the dimension of
 (A) frequency (B) velocity (C) angular momentum (D) time
90. Given that $y = A \sin \left[\left(\frac{2\pi}{\lambda} (ct - x) \right) \right]$, where y and x are measured in metres. Which of the following statements is true?
 (A) The unit of λ is same as that of x and A (B) The unit of λ is same as that of x but not of A

- (C) The unit of c is same as that of $\frac{2\pi}{\lambda}$ (D) The unit of $(ct-x)$ is same as that of $\frac{2\pi}{\lambda}$
91. Which of the following sets cannot enter into the list of fundamental quantities in any system of units?
- (A) length, mass and density (B) length, time and velocity
(C) mass, time and velocity (D) length, time and mass
92. In the formula $X = 3Y Z^2$, X and Z have dimensions of capacitance and magnetic induction respectively. What are the dimensions of Y in MKSQ system?
- (A) $[M^{-3}L^{-1}T^3Q^4]$ (B) $[M^{-3}L^{-2}T^4Q^4]$ (C) $[M^{-2}L^{-2}T^4Q^4]$ (D) $[M^{-3}L^{-2}T^4Q]$
93. A quantity X is given by $\epsilon_0 L \frac{\Delta V}{\Delta t}$, where ϵ_0 is the permittivity of free space, L is a length, ΔV is a potential difference and Δt is a time interval. The dimensional formula for X is the same as that of
- (A) resistance (B) charge
(C) voltage (D) current
94. In the relation $p = \frac{\alpha}{\beta} e^{-\frac{\alpha Z}{k\theta}}$, p is pressure, Z is distance, k is Boltzmann constant and θ is the temperature. The dimensional formula of β will be
- (A) $[M^0L^2T^0]$ (B) $[ML^2T]$
(C) $[ML^0T^{-1}]$ (D) $[M^0L^2T^{-1}]$
95. The dimensions of the quantities in one (or more) of the following pairs are the same. Identify the pair (s).
- (A) Torque and work (B) Angular momentum and work
(C) Energy and Young's modulus (D) Light year and wavelength
96. The pairs of physical quantities that have the same dimensions is (are)
- (A) Reynolds number and coefficient of friction
(B) Curie and frequency of a light wave
(C) Latent heat and gravitational potential
(D) Planck's constant and torque
97. The SI unit of the inductance, the henry can be written as

- (A) weber/ampere (B) volt-second/ampere
(C) joule / (ampere)² (D) ohm-second

98. Let $[\epsilon_0]$ denote the dimensional formula of the permittivity of the vacuum and $[\mu_0]$ that of the permeability of the vacuum. If M = mass, L = length, T = time and I = electric current.

- (A) $[\epsilon_0] = [M^{-1}L^{-3}T^2I]$ (B) $[\epsilon_0] = [M^{-1}L^{-3}T^4I^2]$
(C) $[\mu_0] = [MLT^{-2}I^{-2}]$ (D) $[\mu_0] = [ML^2T^{-1}I]$

99. L, C and R represent the physical quantities inductance, capacitance and resistance respectively. The combinations which have the dimensions of frequency are

- (A) $\frac{1}{RC}$ (B) $\frac{R}{L}$ (C) $\frac{1}{\sqrt{LC}}$ (D) $\frac{C}{L}$

100. Match the two columns.

Column I	Column II
(A) Boltzmann constant	(p) $[ML^2T^{-1}]$
(B) Coefficient of viscosity	(q) $[ML^{-1}T^{-1}]$
(C) Planck constant	(r) $[MLT^{-3}K^{-1}]$
(D) Thermal conductivity	(s) $[ML^2T^{-2}K^{-1}]$

- A) (A) $\rightarrow$ q, (B) $\rightarrow$ s, (C) $\rightarrow$ r, (D) $\rightarrow$ p B) (A) $\rightarrow$ r, (B) $\rightarrow$ q, (C) $\rightarrow$ p, (D) $\rightarrow$ s
C) (A) $\rightarrow$ s, (B) $\rightarrow$ q, (C) $\rightarrow$ r, (D) $\rightarrow$ p D) (A) $\rightarrow$ s, (B) $\rightarrow$ q, (C) $\rightarrow$ p, (D) $\rightarrow$ r

101. Match the physical quantities given in Column I with dimensions expressed in terms of mass (M), length (L), time (T), and charge (Q) given in Column II.

Column I	Column II
(A) Angular momentum	(p) $[ML^2T^{-2}]$
(B) Latent heat	(q) $[ML^2Q^{-2}]$
(C) Torque	(r) $[ML^2T^{-1}]$
(D) Capacitance	(s) $[ML^3T^{-1}Q^{-2}]$
(E) Inductance	(t) $[M^{-1}L^{-2}T^2Q^2]$
(F) Resistivity	(u) $[L^2T^{-2}]$

(A) (A) $\rightarrow$ s, (B) $\rightarrow$ p, (C) $\rightarrow$ u, (D) $\rightarrow$ t, (E) $\rightarrow$ q, (F) $\rightarrow$ s

(B) (A) $\rightarrow$ r, (B) $\rightarrow$ u, (C) $\rightarrow$ p, (D) $\rightarrow$ t, (E) $\rightarrow$ q, (F) $\rightarrow$ s

(C) (A) $\rightarrow$ t, (B) $\rightarrow$ r, (C) $\rightarrow$ s, (D) $\rightarrow$ q, (E) $\rightarrow$ p, (F) $\rightarrow$ u

(D) (A) $\rightarrow$ q, (B) $\rightarrow$ s, (C) $\rightarrow$ r, (D) $\rightarrow$ p, (E) $\rightarrow$ t, (F) $\rightarrow$ u

102. Column I gives three physical quantities. Select the appropriate units for the choices given in Column II. Some of the physical quantities may have more than one choice.

Column I	Column II
(A) Capacitance	(p) ohm-second
(B) Inductance	(q) coulomb ² – joule ⁻¹
	(r) coulomb (volt) ⁻¹ ,
(C) Magnetic induction	(s) newton (ampere metre) ⁻¹ ,
	(t) volt-second (ampere) ⁻¹

(A) (A) $\rightarrow$ q, s, (B) $\rightarrow$ q, t, (C) $\rightarrow$ t

(B) (A) $\rightarrow$ s, t, (B) $\rightarrow$ q, r, (C) $\rightarrow$ p

(C) (a) $\rightarrow$ q, r, (b) $\rightarrow$ p, t, (c) $\rightarrow$ s

(D) (A) $\rightarrow$ p, r, (B) $\rightarrow$ q, t, (C) $\rightarrow$ s

103. Some physical quantities are given in Column I and some possible SI units in which these quantities may be expressed are given in Column II. Match the physical quantities in Column I with the units in Column II.

Column I	Column II
(A) GM_eM_s G – universal gravitational constant, M_e – mass of the earth, M_s – mass of the sun.	(p) (volt) (coulomb) (metre)
(b) $\frac{3RT}{M}$ R – universal gas constant,	(q) (kilogram (metre) ³ (sec ond) ⁻²

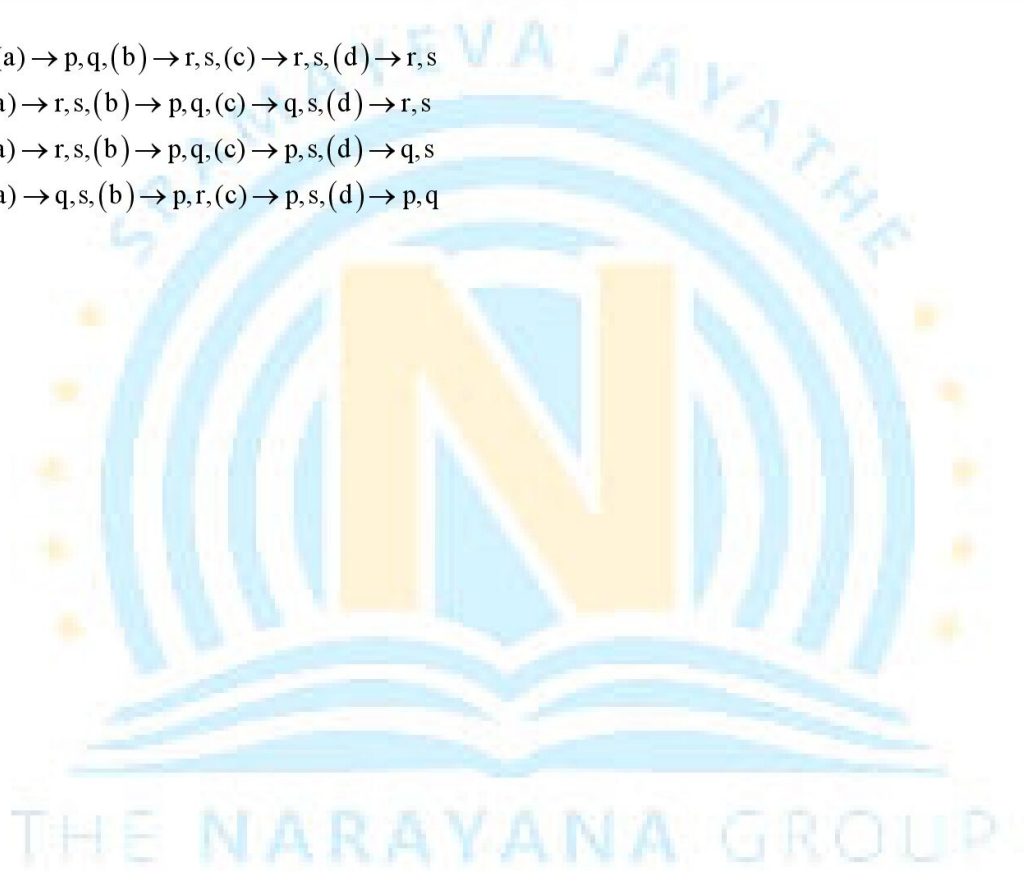
T-absolute temperature, M-molar mass.	
(C) $\frac{F^2}{q^2 B^2}$ F-force, q-charge, B-magnetic field.	(r) $(\text{metre})^2 (\text{second})^{-2}$
(d) $\frac{GM_e}{R_e}$ G-universal gravitational constant, M_e - mass of the earth, R_e -radius of the earth.	(s) $(\text{farad}) (\text{volt})^2 (\text{kg})^{-1}$

(A) (a) $\rightarrow$ p, q, (b) $\rightarrow$ r, s, (c) $\rightarrow$ r, s, (d) $\rightarrow$ r, s

(B) (a) $\rightarrow$ r, s, (b) $\rightarrow$ p, q, (c) $\rightarrow$ q, s, (d) $\rightarrow$ r, s

(C) (a) $\rightarrow$ r, s, (b) $\rightarrow$ p, q, (c) $\rightarrow$ p, s, (d) $\rightarrow$ q, s

(D) (a) $\rightarrow$ q, s, (b) $\rightarrow$ p, r, (c) $\rightarrow$ p, s, (d) $\rightarrow$ p, q



KEY & SOLUTIONS

1	A	2	C	3	300	4	15	5	500
6	4	7	6	8	1	9	D	10	B
11	D	12	C	13	AD	14	AD	15	ACD
16	CD	17	AC	18	ABC	19	B	20	A
21	C	22	C	23	A	24	D	25	A
26	B	27	49	28	562	29	3	30	6
31	A	32	D	33	C	34	B	35	C
36	C	37	A	38	B	39	BC	40	BD
41	B	42	A	43	AD	44	BD	45	ABCD
46	AB	47	ABD	48	ABC	49	A	50	2
51	2	52	3	53	2	54	2	55	B
56	A	57	A	58	B	59	C	60	D
61	3	62	2	63	2	64	2	65	3
66	A	67	ACD	68	ABD	69	3	70	AC
71	C	72	AB	73	B	74	D	75	B
76	A	77	B	78	D	79	D	80	C
81	C	82	A	83	D	84	B	85	C
86	C	87	C	88	A	89	A	90	A
91	B	92	B	93	D	94	A	95	AD
96	ABC	97	ABCD	98	BC	99	ABC	100	D
101	B	102	C	103	A	104		105	

SOLUTIONS

1. Strain = $\ell_0 (1 + \alpha \Delta T) - (\ell_0 + x)$ Here x = compression in springs.

$$\text{Stress} = S = Y + \text{strain}$$

$$\Rightarrow S = \frac{Y\ell_0(1 + \alpha \Delta T) - Y(\ell_0 + x)}{\ell_0} = \frac{Y\ell_0 \alpha \Delta T - Yx}{\ell_0}$$

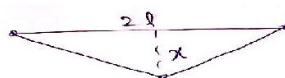
$$\text{But } S = \frac{Kx}{A} \Rightarrow x = \frac{SA}{k}$$

$$\Rightarrow S = \frac{\left(Y\ell_0 \alpha \Delta T - \frac{YSA}{K} \right)}{\ell_0}$$

Solve to get S .

2. Change in length = $S\ell$.

$$S\ell = 2\sqrt{x^2 + \ell^2} - 2\ell$$

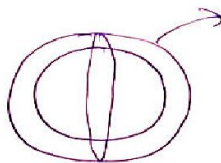


$$S\ell = 2\ell \left[\left(1 + \left(\frac{x}{\ell} \right)^2 \right)^{\frac{1}{2}} - 1 \right]$$

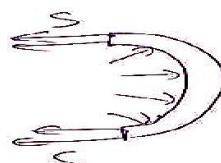
$$\text{Strain} = \frac{S\ell}{2\ell} = \left(1 + \frac{x^2}{\ell^2} \right)^{\frac{1}{2}} - 1 \approx \frac{x^2}{2\ell^2} [x \ll \ell]$$

$$3. \quad \frac{\Delta V}{V} = \frac{SP}{B} \quad \text{and} \quad \frac{\Delta V}{V} = r \Delta T$$

$$SP = \text{Excess pressure} = Br \Delta T$$



F.B.D of right half



$t \equiv$ thickness

$$(SP)\pi R^2 = S \times 2\pi R \times t$$

$$S = \frac{R(SP)}{2t}$$

$$4. \quad V = \pi R^2 h$$

$$\frac{dV}{V} = \frac{2dR}{R} + \frac{dh}{h} \Rightarrow \frac{dh}{h} = \frac{dV}{V} - \frac{2dR}{R}$$

$$\text{Stress} = Y \frac{dh}{h}$$

$$10. \quad \text{Excess pressure} = \frac{mg}{A}$$

$$SP = \left(\frac{dV}{V} \right) K \quad V = \frac{4}{3} \pi R^3$$

$$\frac{dV}{V} = \frac{3dR}{R}$$

$$\frac{mg}{A} = 3K \frac{dR}{R} \Rightarrow \frac{dR}{R} = \frac{mg}{3KA}$$

$$11. \quad u = \frac{1}{2} kx^2 \quad F = kx$$

$$u = \frac{1}{2} Fx \quad \text{Put } F = 2\Omega N$$

$$x = 1 \text{ mm}$$

$$12. \quad \text{Stress} = \frac{F}{bc} \quad \text{strain} = \frac{F}{\eta bc}$$

$$\phi = \text{strain} = \frac{F}{\eta bc}$$

$$20. \quad \text{From screw gauge z.e.} = +0.20 \text{ mm}$$

$$\text{Dia of chalk} = 3.5 - 0.2 = 3.3 \text{ mm} = 0.33 \text{ cm}$$

$$\text{For vernier callipers, z.e.} = -0.05 \text{ cm}$$

$$\text{Thus it must show } 0.33 - 0.05 = 0.28 \text{ cm}$$

So (a) is correct

21. Radiation correction refers to correction introduced in the Joule's calorimeter due to adiation losses. End correction corresponds to the meter bridge. Backlash error is present in the apparatus involving screw system.

Index error corresponds to the optical bench.

22. $a + b = 14 \pm 0.14 \Rightarrow \%error = \frac{0.14}{14} \times 100 = 1\%$

$$a - b = 2 \pm 0.14 \Rightarrow \%error = \frac{0.14}{2} \times 100 = 7\%$$

$$a \times b = 48 \pm 0.96 \Rightarrow \%error = \frac{0.96}{48} \times 100 = 2\%$$

Order of % error $x < z < y$

23. $Y = \frac{F\ell}{A\Delta\ell}$

$$\Delta\ell = \frac{1}{4} mm = 0.25 \times 10^{-3} m$$

$$F = 50 \times 10 N = 500 N$$

$$\ell = 3 m$$

$$A = 10^{-5} m^2$$

$$\therefore Y = \frac{5 \times 10^2 \times 3}{10^{-5} \times 0.25 \times 10^{-3}} = 6 \times 10^{11} N / m^2$$

24. Effective length of simple pendulum
 $= 63.5 + 1.55 = 65.05 = 65.1 \text{ cm}$

25. $A = \ell \times b$

$$\therefore \frac{\Delta A}{A} = \frac{\Delta \ell}{\ell} + \frac{\Delta b}{b}$$

$$\Rightarrow \Delta A = \left(\frac{\Delta \ell}{\ell} + \frac{\Delta b}{b} \right) \times A = 0.2 cm^2$$

27. $g = \frac{4\pi^2 L}{T^2}$

Here, $T = \frac{t}{n}$ and $\Delta T = \frac{\Delta t}{n}$. Therefore,

$$\frac{\Delta T}{T} = \frac{\Delta t}{t}$$

The errors in both L and t are the least count errors.

Therefore,

$$\left(\frac{\Delta g}{g} \right) = \left(\frac{\Delta L}{L} \right) + 2 \left(\frac{\Delta T}{T} \right)$$

$$= \frac{0.1}{20.0} + 2 \left(\frac{1}{90} \right) = 0.027$$

Thus, the percentage error in g is

$$100\left(\frac{\Delta g}{g}\right) = 100\left(\frac{\Delta L}{L}\right) + 2 \times 100\left(\frac{\Delta T}{T}\right)$$

28. Least count of screw gauge = $\frac{1}{50} \text{ mm} = 0.02 \text{ mm}$

Negative zero error = $(-1 + 44 \times 0.02)$

= -0.12 mm

Reading = $5 \times 1 \text{ mm} + 25 \times 0.02 \text{ mm} + 0.12 \text{ mm}$

= $5.62 \text{ mm} = 562 \times 10^{-5} \text{ m}$

29. Least count = $\frac{1 \text{ mm}}{10} = 0.1 \text{ mm}$; Zero error

= $-(10 - 6) \times 0.1 = -0.4 \text{ mm}$;

Reading = $6 + 5 \times (0.1) - (-0.4) = 6.9 \text{ mm}$

30. $R = \frac{V}{I} = \frac{10^3}{1.43} = 700 \Omega = 7 \times 10^{-1} \text{ k}\Omega$

43. The slope of the linear portion of the curve gives the Young's modulus of the material. The slope of the linear portion OP for material A is greater than that of the linear portion Or for material B. Hence, statement (a) is correct. The plastic region for material A (from P to Q) is greater than that (from R to S) for material B, which indicates that material A is more ductile. Hence, statement (b) is correct. The breaking stress for material B (i.e., stress corresponding to point S) is less than that for material A (i.e., stress corresponding to point Q), Which implies that material B is more brittle. Hence choice (c) is also incorrect. Material A is stronger than material B because it can withstand a greater stress before it breaks. The breaking stress is the stress corresponding to point Q for material A and to point S for material B. Hence, the correct choices are (a) and (d).

44. Elongation = $= 2\sqrt{l^2 + d^2} - 2l$

$$= 2 \left[l^2 \left(1 + \frac{d^2}{l^2} \right) \right]^{\frac{1}{2}} - 2l$$

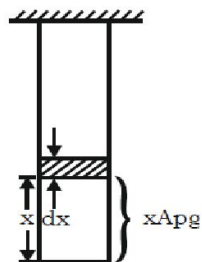
$$= 2 \left[1 + \frac{d^2}{2l^2} \right] - 2l$$

$$= \frac{d^2}{l}$$

$$\therefore \text{Strain } \epsilon = \frac{d^2}{2l^2}$$

$$\text{Tension} = \gamma a \left(\frac{d^2}{2l^2} \right)$$

45. Consider an element as shown in the figure.



$$\text{Stress in the element} = \frac{\text{Force}}{\text{Area}} = \frac{xAp g}{A} = xpg$$

Now, elastic potential energy stored in the wire is

$$dU = \frac{1}{2} (\text{Stress}) (\text{Strain}) (\text{Volume})$$

$$= \frac{1}{2} \cdot \frac{(\text{Stress})^2}{Y} (\text{Volume})$$

$$dU = \frac{1}{2} \cdot \frac{(xpg)^2}{Y} A dx = \frac{1}{2} \cdot \frac{\rho^2 g^2 A}{Y} x^2 dx$$

$$\text{Total elastic potential energy} = \frac{1}{2} \cdot \frac{\rho^2 g^2 A}{Y} \int_0^L x^2 dx$$

$$= \frac{\rho^2 g^2 AL^3}{6Y}$$

46. Tension in the rod at a point $\frac{l}{4}$ from the centre $T = \int_{l/4}^{l/2} \frac{m}{l} x \omega^2 dx = \frac{3}{32} m \omega^2 l$

$$\text{Elongation in the rod } \Delta l = \frac{m \omega^2}{2AY} \int_{-l/2}^{l/2} \left(x^2 - \frac{l^2}{4} \right) dx = \frac{m \omega^2 l^2}{12AY}$$

47. Tensile stress $= \frac{F \cos \theta}{A'}$ where $A' = \frac{A}{\cos \theta}$

$$\text{Shear stress} = \frac{F \cos \theta}{A'}$$

48. $2T \sin \frac{d\theta}{2} = \frac{N}{\sqrt{2}} = dm g$

$$T d\theta = \frac{m}{2\pi R} (R d\theta) g$$

$$T = \frac{mg}{2\pi}$$

$$\text{Elongation} = \left(\frac{T}{aY} \right) 2\pi R = \frac{mgR}{aY}$$

$$\text{Energy} = \frac{1}{2} \times \text{stress} \times \text{strain} = \frac{m^2 g^2}{8\pi^2 a^2 Y}$$

49. Given $YA = l$

$$\text{Since } e = \frac{F.l}{YA}$$

$$\Rightarrow F = e$$

$\therefore x$ and y represents F & Δl

$$\Delta U = \frac{1}{2} Fe = \frac{1}{2} \left(\frac{YAe}{l} \right) e = \frac{e^2}{2}$$

For a given material Y is constant

$$50. \quad \Delta \ell = \frac{F \ell}{AY} \quad \frac{\Delta \ell}{(F/A)} = \frac{\ell}{Y} = \text{slope of curve}$$

$$\frac{\ell}{Y} = \frac{(4-2) \times 10^{-3}}{4000 \times 10^3}$$

$$\text{Given } \ell = 1m \rightarrow Y = \frac{4000 \times 10^3}{2 \times 10^3} = 2 \times 10^9 N/m^2$$

$$51. \quad \text{Shear strain} = \frac{\Delta x}{Y}$$

$$52. \quad \text{Increase in pressure is } \Delta p = \frac{Mg}{A}$$

$$\text{Bulk modulus is } B = \frac{\Delta p}{(\Delta V/V)}$$

$$\therefore \frac{\Delta V}{V} = \frac{\Delta p}{B} = \frac{Mg}{AB} \dots (i)$$

$$\text{Also, the volume of the sphere is } V = \frac{4}{3} \pi R^3 \quad p \frac{\Delta V}{V} = \frac{3 \Delta R}{R}$$

$$\text{Or } \frac{\Delta R}{R} = \frac{1}{3} \cdot \frac{\Delta V}{V},$$

Using Eq. (i) we get

$$\frac{\Delta R}{R} = \frac{Mg}{3AB} \Rightarrow \therefore \alpha = 3$$

$$53. \quad U = K e^2 \text{ and}$$

$$U = \frac{1}{2} Y \left(\frac{e}{l} \right)^2 V$$

$$54. \quad 2T \sin \frac{\Delta \theta}{2} = dm \omega^2 r$$

$$2T \left(\frac{\Delta \theta}{2} \right) = \rho \times A \times r^2 \omega^2$$

$$\sigma = \frac{T}{A} = \rho r^2 \omega^2$$

$$\omega = \frac{1}{r} \sqrt{\frac{\sigma}{\rho}} = 2 \text{ rad/s}$$

$$56. \quad \Delta V = \frac{FL}{Y} (1 - 2\mu)$$

$$\frac{\Delta V}{V} = \frac{FL}{AY} (1 - 2\mu)$$

$$58. \quad \text{The tension is not constant along the length of the plank. Let force at a distance } x \text{ from one end be } F.$$

Then $\frac{F}{x} = \frac{F_0}{l}$

Or $F = \frac{F_0 x}{l}$

Stress at any cross-section

$$\sigma = \frac{F}{A} = Y \times \text{strain}$$

$$= \frac{F_0 x}{lA} \times \frac{\text{change in length of element } dx}{dx}$$

Increase in length of element,

$$dx = \frac{F_0}{lAY} x dx$$

$$\Delta l = \frac{F_0}{lAY} \int_0^l x dx$$

$$= \frac{1}{2} \frac{F_0}{AY} l^2$$

Hence strain $\frac{\Delta l}{l} = \frac{1}{2} \frac{F_0}{AY}$

61. $\lambda \propto \frac{1}{\sqrt{T}}$. Hence $\frac{\lambda'}{\lambda} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{927+273}{27+273}} = 2$.

$\therefore \lambda' = 2\lambda$. Hence the correct choice is (3).

62. Orbital speed $v_0 \propto \frac{1}{n}$. Therefore, $n = 2$. Now $r_n \propto n^2$. Hence the radius of the new orbit =

$$(2)^2 \times r = 4r.$$

63. $eV_0 = h(\nu - \nu_0)$ Slope $\frac{V_0}{\nu - \nu_0} = \frac{h}{e}$.

64. We know that $r_n \propto n^2$ or $n \propto \sqrt{r_n}$. If r_1 and r_2 are the radii of the initial and final states

Respectively, then $\frac{n_2}{n_1} = \sqrt{\frac{r_2}{r_1}} = \sqrt{\frac{2.12}{0.53}} = 2$.

65. Energy released in annihilation = $0.51 + 0.51 = 1.02$ MeV.

Initial energy = $1 + 1 = 2$ MeV. Therefore, the energy of the two photons is = $1.02 + 2 = 3.02$ MeV.

Hence energy of each photon is $E = 1.51$ MeV. Now, according to Duane – Hunt law, the wavelength of a photon of energy E (in eV) is given by

$$\lambda = \frac{hc}{E} = \frac{12400 \text{ eV } \text{\AA}}{1.51 \times 10^6 \text{ eV}} = 8.2 \times 10^{-3} \text{ \AA} \text{ which is choice (c).}$$

66. $R = \frac{V}{9} \Rightarrow \frac{dR}{R} = \frac{dV}{V} + \frac{di}{i}$

$$dR\% = \left(\frac{0.5}{8} + \frac{0.2}{2} \right) 100 = 16.25\%$$

$$R = \frac{8}{2} = 4$$

$$R = (4 \pm 16.25) \Omega$$

67. Conceptual

$$68. \therefore R_A = \frac{1}{0.02}$$

$$500 \Omega$$

$$69. P = \frac{W_a}{W_a - W_b} \Rightarrow m\rho = mW_a - m(W_a - W_b)$$

$$\frac{d\rho}{\rho} = \frac{dW_a}{W_a} - \frac{d(W_a - W_b)}{W_a - W_b}$$

$$\left(\frac{d\rho}{\rho} \right)_{\max} = \frac{-dW_a}{W_a} + \frac{dW_a}{W_a - W_b} + \frac{dW_b}{W_a - W_b}$$

$$= \frac{-0.1}{10} + \frac{0.1}{5} + \frac{0.1}{5} = \frac{0.3}{10} \text{ or } 3\%$$

$$70. 1MSD = \frac{1}{8}; 5VSD = 4MSD$$

$$1VSD = \frac{4}{5}MSD = \frac{4}{5} \times \frac{1}{8} = \frac{1}{10} \text{ cm}$$

least count of vernier callipers

$$= 1MSD - 1VSD$$

$$= \frac{1}{8} \text{ cm} - \frac{1}{10} \text{ cm} = 0.025$$

Pitch of screw gauge = 0.05 cm

$$\text{Least count of screw gauge} = \frac{0.05}{100} \text{ cm} = 0.005 \text{ mm}$$

Least count of linear scale of screw gauge = 0.05 cm

$$\text{Pitch} = 0.05 \times 2 = 0.1 \text{ cm}$$

$$\text{Least count of screw gauge} = \frac{0.1}{100}$$

$$= 0.01 \text{ mm}$$

For screw gauge

$$\text{One division of ms} = \frac{1}{8} \text{ cm}$$

Let the pitch = y

$$\text{Pitch} = y = \frac{2}{8} \text{ cm} = \frac{1}{4} \text{ cm} = 0.25 \text{ cm}$$

71. Hence $1MSD = X; N.V.S.D = (N+1)MSD$

$$1VSD = \frac{N+1}{N}MSD$$

72. Conceptual

$$73. LC = \frac{1}{50} = 0.02 \text{ mm}$$

74. The instrument has positive zero error

$$e = +3 \times 0.02 = 0.06 \text{ mm}$$



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